



KTM650

Carbon based Composite - Heat Spreader

KEY PROPERTIES

- Extremely high thermal conductivity
- Low density
- Excellent machinability
- CTE matching with silicon
- Suitable for complex geometries

APPLICATIONS

- Diode lasers
- Power transistor substrates
- LED substrates
- Wireless communications
- Electrical drives
- Aerospace power supplies
- Medical electronic systems
- Avionics systems

MAIN PROPERTIES

Property	Value	Unit
Density	2.57	g/cm³
Flexural strength (XY/Z)	102 / 16.9	MPa
Young Modulus (XY/Z)	69.7 / 5.5	GPa
Thermal Conductivity (XY)	650	W/m·K
Thermal Conductivity (Z)	45	W/m·K
Thermal Diffusivity (XY)	390	mm²/s
Thermal Diffusivity (Z)	27	mm²/s
CTE average	6.5	x10 ⁻⁶ K ⁻¹
CTE (XY/Z)	2.4 / 14.7	x10 ⁻⁶ K ⁻¹
Specific Heat	0.65	J/g·K
Electrical Conductivity	0.8	MS/m

* All properties measured at 20°C unless otherwise stated.

** Measured in a wall thickness of 2 mm

OPERATING CONDITIONS

Max working temp (air)	400 °C
Recomended temp	<350 °C
Continuous temp	300 °C

ELECTRICAL BEHAVIOR

Conductive	
Electrical Conductivity	0.8 MS/m

MATERIAL INFORMATION

Material type	Carbon-based composite
Structure	Anisotropic composite
Key feature	Thermal management
Machinability	Excellent
Color	Black

Local overheating may occur at electrical contact points due to current concentration and geometry effects.

The information provided is for guidance only and does not constitute a guarantee.

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